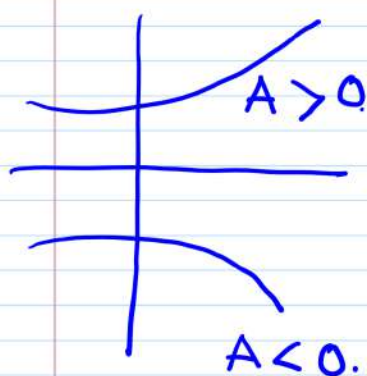


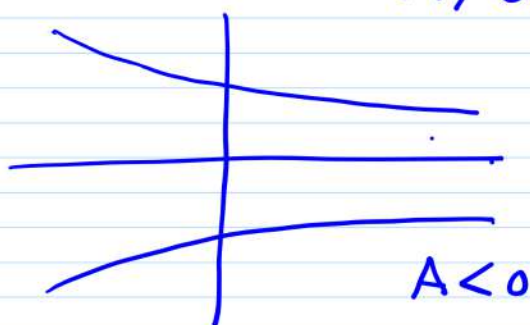
Ordinary differential equations (ODEs)

$$a > 0$$

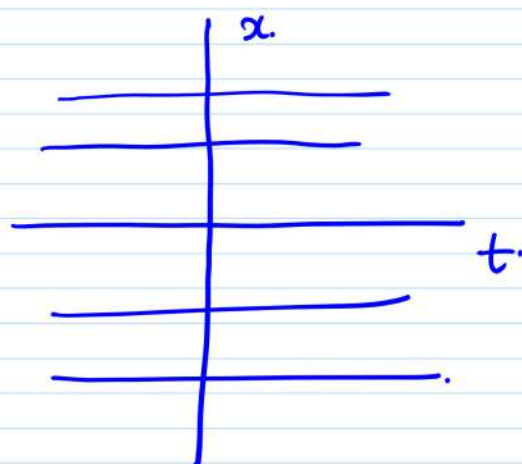


$$a < 0$$

$$A > 0$$



$$a = 0$$



Example.

$$\ddot{x} + p^2 x = 0; \quad \ddot{x} = -p^2 x.$$

SHM. $x = A \sin pt + B \cos pt$ (??)

Definition. x is independent variable.

An ordinary differential equation. is. an. equation. that involves. one or several. derivatives of $y = y(x), y^{(1)}, \dots, y^{(n)}$,

i.e., $F(x, y, y^{(1)}, \dots, y^{(n)}) = 0$. — (1).

(1) is called n -th order ODE.

§ First order ODE :

→ (2).

$$\begin{cases} F(x, y, y^{(1)}) = 0 \rightarrow \text{implicit form.} \\ y^{(1)} = f(x, y) \rightarrow \text{explicit form.} \end{cases}$$

Example. $\bar{x}^3 y' - 4y^2 = 0$, $(x \neq 0)$ $\rightarrow$ implicit.
 $y' = 4y^2 x^3$ $\rightarrow$ explicit.

§. Concept of Solution.:

A function $h = h(x)$ is called a solution of (2) defined on $a < x < b$, if $h(x)$ is defined and differentiable on $a < x < b$ and $F(x, h(x), h'(x)) = 0$.

The curve $h(x)$ is called solution curve of (2).

Example.: $\frac{dy}{dx} = \cos x$.

$$y = \sin x + C.$$

family of solution curves.

Initial value Problem.:

$$y' = f(x, y), \quad y(x_0) = y_0.$$

This is called initial value problem.

Example. $\frac{dy}{dx} = y, \quad y(0) = 1.$

$$y = e^x.$$

Separable. ODEs.

$$\frac{dy}{dx} = f(x, y). \quad \text{--- (3)}$$

and if $f(x, y) = F(x) G(y)$.

(3) is called Separable ODE.

$$\frac{dy}{dx} = F(x) G(y).$$

$$\frac{dy}{G(y)} = F(x) dx;$$

$$c. + \int \frac{dy}{G(y)} = \int F(x) dx, \quad \text{if } F, \frac{1}{G} \text{ are continuous.}$$

constant.

Example.

$$\frac{dy}{dx} = 1 + y^2.$$

$$\frac{dy}{1+y^2} = dx.$$

$$y = \tan(x+c). \quad !..$$

§. Reduction. to Separable form.

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right). \quad \text{--- (4)}$$

$$\frac{y}{x} = v, \quad \frac{dy}{dx} = \frac{dv}{dx} x + v.$$

$$x \frac{dv}{dx} = f(v) - v.$$

$$\frac{dv}{f(v)-v} = \frac{dx}{x}.$$

$$\int \frac{dv}{f(v)-v} = \int \frac{dx}{x} + c,$$

if $f(v)-v \neq 0$
and $\frac{1}{f(v)-v}$ is
continuous.

Example: $2xy y' = y^2 - x^2.$

$$y' = \frac{y^2 - x^2}{2xy}$$

$$\frac{y}{x} = v.$$

$$x \cdot \frac{dv}{dx} + v = \frac{v^2 - 1}{2v}.$$

$$x \cdot \frac{dv}{dx} = -\frac{1+v^2}{2v}.$$

$$\frac{2v \cdot dv}{1+v^2} = -\frac{dx}{x}.$$

$$\log(1+v^2) + \log x = \log |c|.$$

$$\Rightarrow x(1+v^2) = c.$$

$$\Rightarrow (x^2 + y^2) = cx.$$

$$\Rightarrow \left(x - \frac{c}{2}\right)^2 + y^2 = \left(\frac{c}{2}\right)^2.$$

//

§.

$$u(x, y) = c.$$

$$du = 0$$

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0.$$

$$\frac{dy}{dx} = - \frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = - \frac{M(x, y)}{N(x, y)}.$$

$$M(x, y) dx + N(x, y) dy = 0.$$

$$\Rightarrow M(x, y) = \frac{\partial u}{\partial x}.$$

$$N(x, y) = \frac{\partial u}{\partial y}.$$

$$\Rightarrow \boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}.$$

Lecture-2. Date 11.03.26.

Recall

$$\frac{dy}{dx} = f(x, y). \rightarrow \text{Explicit form.}$$

$$F(x, y, y') = 0 \rightarrow \text{Implicit form.}$$



$$M dx + N dy = 0, \quad \begin{array}{l} M = M(x, y) \\ N = N(x, y). \end{array}$$

— 1)

A first order ODE. 1) is called exact if the form $Mdx + Ndy$ is exact if $u = u(x, y)$ such that.

$$Mdx + Ndy = du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy.$$

$$\Rightarrow M = \frac{\partial u}{\partial x}, \quad N = \frac{\partial u}{\partial y}.$$

$$\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}.$$

$$\Rightarrow \boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}.$$

$\because u$ has continuous 2nd - Partial derivatives

If ① is exact, then $u(x, y) = c$ is a solution of ①

Example.

$$y dx + x dy = 0$$

Since $y dx + x dy = d(xy).$

$\rightarrow$ is exact and $xy = c$ gives solution.

Theorem:- Assume that $M = M(x, y)$ and $N = N(x, y)$ have continuous partial derivatives. Then

$Mdx + Ndy = 0$ is exact if and only if.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$



§. $M dx + N dy = 0$ — ①.

Assume that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

$$\frac{\partial u}{\partial x} = M, \quad \frac{\partial u}{\partial y} = N$$

$$u = \int M dx + k(y).$$

$$N = \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \int M dx + k'(y)$$

$$k'(y) = N - \frac{\partial}{\partial y} \int M dx.$$

$$k(y) = \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy$$

Example. $\underbrace{\cos(x+y)}_{M} dx + \underbrace{(3y^2 + 2y + \cos(x+y))}_{N} dy = 0$ (*)

$$\frac{\partial M}{\partial y} = -\sin(x+y) \quad (*) \text{ is exact.}$$

$$\frac{\partial N}{\partial x} = -\sin(x+y).$$

$$u = u(x, y) \quad \frac{\partial u}{\partial x} = M. \quad \& \quad \frac{\partial u}{\partial y} = N.$$

$$\frac{\partial u}{\partial x} = \cos(x+y).$$

$$u = \sin(x+y) + k(y). \quad \text{--- } (**)$$

$$\frac{\partial u}{\partial y} = \cos(x+y) + k'(y).$$

$$3y^2 + 2y + \cos(x+y) = \cos(x+y) + k'(y).$$

$$k'(y) = 3y^2 + 2y.$$

$$k(y) = y^3 + y^2.$$

$$u = \sin(x+y) + y^3 + y^2.$$

$u(x,y) = c$ is a solution of (*)

Example. $-y dx + x dy = 0$ (Non-exact ODE).
 $\begin{matrix} \parallel & \parallel \\ M & N. \end{matrix}$ ①

$$\frac{\partial M}{\partial y} = -1 \quad / \quad \frac{\partial N}{\partial x} = 1.$$

$$\text{so, } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

① is not exact.

But $\frac{-y dx + x dy}{x^2} = 0$

$$\Rightarrow d\left(\frac{y}{x}\right) = 0.$$

$$\Rightarrow \frac{y}{x} = c. \Rightarrow y = cx.$$

Note: $x=0$ is also a solution of the ODE ①.

§. Reduction to exact ODE.

$$P dx + Q dy = 0, \text{ (not exact)}$$

—(*)

$$\text{i.e., } \frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}.$$

Definition.: If there is a function $F = F(x, y)$ such that

$$FP dx + FQ dy = 0.$$

is exact, then F is called an integrating factor of (*).

Example:- $-y dx + x dy = 0.$

$$F = \frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{xy}, \frac{1}{x^2 + y^2}.$$

§ How to find an integrating factor.

$FP dx + FQ dy = 0$ is exact.

$$\frac{\partial(FP)}{\partial y} = \frac{\partial(FQ)}{\partial x}.$$

$$\Rightarrow \frac{\partial F}{\partial y} P + F \frac{\partial P}{\partial y} = \frac{\partial F}{\partial x} Q + F \frac{\partial Q}{\partial x}.$$

** This is very complicated when F is function of both x and y .

§§. $F = F(x, y)$ is a function of x .

$$F = G(x).$$

$$F \frac{\partial P}{\partial y} = \frac{\partial F}{\partial x} Q + F \frac{\partial Q}{\partial x}.$$

$$G \frac{\partial P}{\partial y} = G' Q + G \frac{\partial Q}{\partial x}.$$

$$\frac{1}{G} dG = \underbrace{\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right)}_{R} dx.$$

$$\ln G = \int R dx.$$

$$F = G = e^{\int R dx}.$$

Similarly, Suppose, $F^* = G(y)$.

$$\frac{\partial G}{\partial y} P + \frac{\partial P}{\partial y} G = G \frac{\partial Q}{\partial x}.$$

$$\frac{dG}{G} = \frac{1}{P} \underbrace{\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)}_{R^*} dy.$$

$$F^* = G = e^{\int R^* dy}.$$

Theorem:- If F is a function of x and R is a function of x , then.

$F = e^{\int R dx}$ is an integrating factor of (*).

Theorem: If F is a function of y and R^* is a function of y , then.

$F^* = e^{\int R^* dy}$ is an integrating factor of $(*)$.

Example. $(\underbrace{e^{x+y} + ye^y}_{\substack{\parallel \\ P.}}) dx + (\underbrace{xe^y - 1}_{Q.}) dy = 0. \quad (*)$

$$\frac{\partial P}{\partial y} = e^{x+y} + ye^y + e^y$$

$$\frac{\partial Q}{\partial x} = e^y.$$

$$R = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = \frac{e^{x+y} + ye^y}{xe^y - 1}$$

$$R^* = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = - \left(\frac{e^{x+y} + ye^y}{e^{x+y} + ye^y} \right) = -1.$$

$$F^* = e^{\int R^* dy} = e^{-y}.$$

$$(\underbrace{e^x + y}_{\substack{\parallel \\ M.}}) dx + (\underbrace{x - e^{-y}}_{\substack{\parallel \\ N.}}) dy = 0.$$

$$\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 1.$$

$$u = u(x, y), \quad \frac{\partial u}{\partial x} = M = e^x + y.$$

$$u = e^x + xy + k(y). \quad \text{---} (**)$$

$$N = \frac{\partial u}{\partial y} = x + k'(y).$$

$$x - e^{-y} = x + k'(y).$$

$$\Rightarrow k'(y) = -e^{-y}.$$

$$\Rightarrow k(y) = e^{-y}$$

$$u = e^x + xy + e^{-y}.$$

Solution $u(x, y) = c. \left[\begin{array}{l} c \text{ is} \\ \text{constant.} \end{array} \right]$

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Lecture-3 Date 13.03.26.

§. First Order Linear ODEs.

A first order linear ODE is of the form.

$$\frac{dy}{dx} + p(x)y = r(x). \quad \text{---} \textcircled{1}.$$



Standard form.

of first order linear ODE.

ODE is defined on.
interval $I \subseteq \mathbb{R}$.

① is called homogeneous if $r(x) = 0 \cdot \forall x \in I$.
And if $r(x) \neq 0$ on I , then (1) is called non-homogeneous.

Example. $\cos x y' + y \sin x = x \quad \text{--- (1)}$.

Linear ODE.

but not in the Standard form.

Standard form $y' + y \tan x = x \sec x$.

homogeneous: $\frac{dy}{dx} + p(x)y = 0 \quad \text{--- (*)}$

$$\frac{dy}{y} = -p(x) dx.$$

$$\Rightarrow \ln y = \int -p(x) dx + c.$$

$$\Rightarrow y = c_1 e^{-\int p(x) dx} \quad . \quad c_1 \text{ is constant.}$$

Observe that: $y \equiv 0$ is also a solution of (*)
[$\underbrace{y(x) = 0 \quad \forall x \in I}$].

General situation.

$$\frac{dy}{dx} + p(x)y = r(x).$$

$$\underbrace{(p(x)y - r(x))}_{P} dx + \underbrace{1}_{Q} dy = 0 \quad \text{--- (1)}.$$

$$\frac{\partial P}{\partial y} = p(x), \quad \frac{\partial Q}{\partial x} = 0.$$

(1) is not exact.

$$R = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = p(x).$$

$$R^* = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = \frac{-p(x)}{P}.$$

$$F = G(x) = e^{\int R dx} = e^{\int p(x) dx}.$$

Put $\int p(x) dx = h.$

$$e^h \frac{dy}{dx} + e^h p(x) y = e^h r(x).$$

$$\frac{d}{dx} (e^h \cdot y) = e^h r(x).$$

$$e^h y = \int e^h r(x) dx + c.$$

$$y = e^{-h} \left(\int e^h r(x) dx + c \right) \rightarrow \text{constant}.$$

if $r(x) = 0$, $y(x) = e^{-h(x)} \cdot c.$

Reduction to linear ODE: Bernoulli equation:

$$\frac{dy}{dx} + p(x)y = g(x)y^a, \quad \underline{a \in \mathbb{R}} \quad (*)$$

If $a=0$, or 1 , (*) is linear ODE.

$a \neq 0, 1$.

$$\frac{dy}{dx} + p(x)y = g(x)y^a.$$

$$\frac{1}{y^a} \frac{dy}{dx} + p(x)y^{1-a} = g(x).$$

Put

$$u = y^{1-a}.$$

$$u' = (1-a)y^{-a}y'.$$

$$u' + (1-a)p(x)u = (1-a)g(x).$$

Linear ODE.

Logistic equation:

$$\frac{dy}{dx} = Ay - By^2, \quad A, B \text{ are constants.}$$

$$\frac{dy}{dx} - Ay = -By^2.$$

$$\frac{1}{y^2} \frac{dy}{dx} - \frac{A}{y} = -B.$$

$$\text{Set } u = 1/y. \quad u' = -\frac{1}{y^2}y'.$$

$$u' + Au = B.$$

$$F = e^{Ax}, \quad u = e^{-Ax} \left(\int e^{Ax} B dx + c \right)$$

$$u = e^{-Ax} \left(\frac{B}{A} e^{Ax} + c \right) \\ = \left(\frac{B}{A} + c e^{-Ax} \right)$$

$$y = \frac{1}{u} = \frac{1}{\left(\frac{B}{A} + c e^{-Ax} \right)} ; c \text{ is constant.}$$

Existence and uniqueness of Solution for IVP.

$$\frac{dy}{dx} = f(x, y). \quad y(x_0) = y_0. \quad \text{--- (1)}$$

Question: When (1) has a Solution.

If Solution exists, Is this unique?

Example 1: $\frac{dy}{dx} = x, \quad y(0) = 1.$

$$y = \frac{x^2}{2} + 1.$$

is a unique solution.

Example 2: $x \frac{dy}{dx} = y - 1, \quad y(0) = 1.$

$$y - 1 = cx, \quad c \text{ is an arbitrary constant.}$$

Infinitely many solutions.

Example 3: $\left| \frac{dy}{dx} \right| + |y| = 0$, $y(0) = 1$.
has no solution.

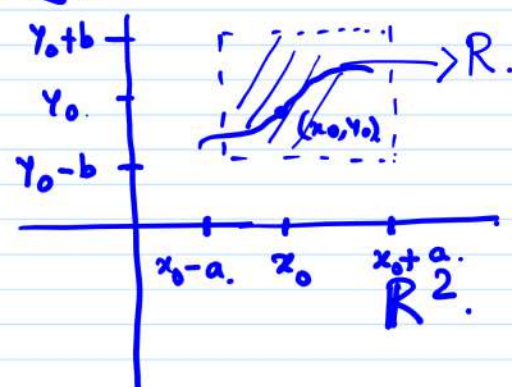
Remark 1. An IVP may have unique solution, infinitely many solutions, no solution.

Problem a) Under what condition (1) has at least one solution.

(b) ——— | ———. (1) has a unique solution.

Theorem 1 [Picard's Existence theorem].

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0. \quad (1)$$



If $f(x, y)$ is continuous in

R , : $|x - x_0| < a$, $|y - y_0| < b$.

and $f(x, y)$ is bounded in R ,

(i.e., $\exists K \in \mathbb{R}_{>0}$ such that $|f(x, y)| \leq K, \forall (x, y) \in R$)

Then (1) has a solution $y = y(x)$.

defined on $|x - x_0| < \alpha$, $\alpha = \min \left\{ a, \frac{b}{K} \right\}$.

Theorem: (Picard's uniqueness theorem).

If $f(x,y)$ and $\frac{\partial f}{\partial y}$ are continuous in the region $R: |x-x_0| < a, |y-y_0| < b$.

and both $f(x,y)$ and $\frac{\partial f}{\partial y}$ are bounded in R , i.e., $|f(x,y)| \leq K \quad \forall (x,y) \in R$.

$$|\frac{\partial f}{\partial y}| \leq M, \quad \forall (x,y) \in R.$$

Then (1) has at most one solution.

By Theorem 1. IVP (1) has a unique solution. and it is defined on.

$$|x-x_0| < \alpha, \quad \alpha = \min \left\{ a, \frac{b}{K} \right\}.$$

Example. $\frac{dy}{dx} = 1+y^2, \quad y(0)=0.$

$$f(x,y) = 1+y^2.$$

$$|x| < 5 = a.$$

$$|y| < 3 = b.$$

$$\frac{\partial f}{\partial y} = 2y.$$

$$|f(x,y)| \leq 10 = K.$$

$$|\frac{\partial f}{\partial y}| \leq 6 = M.$$

$$\alpha = \min \left\{ a, \frac{b}{K} \right\} = \min \left\{ 5, \frac{3}{10} \right\}$$

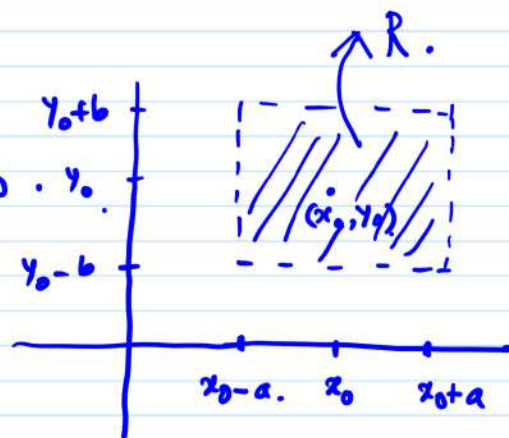
$$|x| < \alpha = \frac{3}{10}.$$

Lecture-4. Date 17.03.26.

Consider IVP:

$$\frac{dy}{dx} = f(x, y). \quad y(x_0) = y_0 \quad (*)$$

$$R: |x - x_0| < a, |y - y_0| < b.$$



Existence:. If $f(x, y)$ is continuous and bounded in R , say $|f(x, y)| \leq K, \forall (x, y) \in R$.
Then $(*)$ has a solution $y = y(x)$ defined in $|x - x_0| < \alpha$, where $\alpha = \min\{a, \frac{b}{K}\}$.

Uniqueness!:- If $f(x, y)$ and $\frac{\partial f}{\partial y}$ both are continuous and bounded in R .

$$\text{say, } |f(x, y)| \leq K, \quad (x, y) \in R.$$

$$|\frac{\partial f}{\partial y}(x, y)| \leq M,$$

then $(*)$ has a solution $y = y(x)$ defined in $|x - x_0| < \alpha$, where $\alpha = \min\{a, \frac{b}{K}\}$.

Discussion:

$$\text{IVP: } \frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

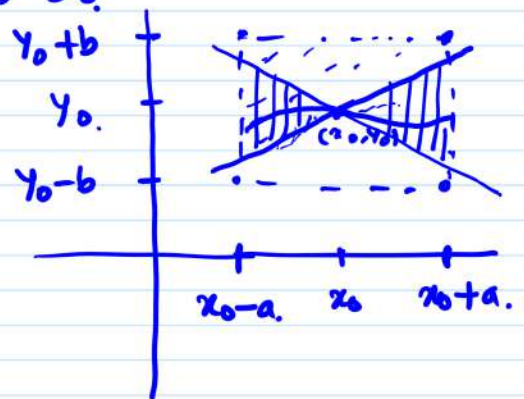
$$\left| \frac{dy}{dx} \right|_{x=x_0} = |f(x_0, y_0)| \leq K.$$

Case-I

$$y - y_0 = (x - x_0) K.$$

$$y - y_0 = aK \leq b.$$

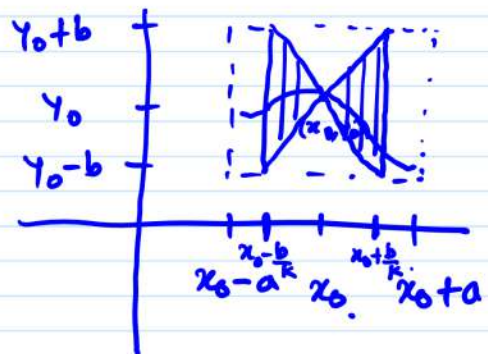
$$\alpha = \min \left\{ a, \frac{b}{K} \right\} = a.$$



Case II

$$y - y_0 = (x - x_0) K.$$

$$\alpha = \min \left\{ a, \frac{b}{K} \right\} = \frac{b}{K}, \quad aK > b.$$



§ Lipschitz's Condition.

Let $f(x, y)$ be defined in R .

$$R: |x - x_0| < a, \quad |y - y_0| < b.$$



A function $f(x, y)$ defined in R satisfies Lipschitz's Condition. if $\exists L > 0$ such that.

$$|f(x, y_1) - f(x, y_2)| \leq L |y_1 - y_2|.$$

$$\forall (x, y_1), (x, y_2) \in R.$$

and L is called. Lipschitz's Constant.

Note:- If $f(x,y)$ is continuous in R such that $\frac{\partial f}{\partial y}$ is continuous and bounded in R .

Then f satisfies Lipschitz's Condition.

Sketch: $|f(x,y_1) - f(x,y_2)| = \left| \frac{\partial f}{\partial y}(p.) \right| |y_1 - y_2|$
 $\leq M |y_1 - y_2|.$

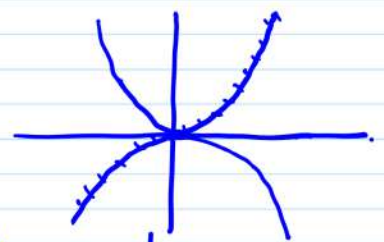
So, f satisfies Lipschitz's Condition. $\forall (x,y_1), (x,y_2) \in R.$

Theorem*: IVP $\frac{dy}{dx} = f(x,y), y(x_0) = y_0 \} (*)$.

Assume that f is continuous and bounded in R , say $|f(x,y)| \leq K \quad \forall (x,y) \in R$, and f satisfies Lipschitz's condition, then.

$(*)$ has a unique solution $y = y(x)$ defined in $|x - x_0| < \alpha, \quad \alpha = \min\left\{a, \frac{b}{K}\right\}.$

Example. $\frac{dy}{dx} = \sqrt{|y|} \quad y(0) = 0.$



$$f(x,y) = \sqrt{|y|}. \quad |f(x,y_1) - f(x,y_2)|$$

$$= \sqrt{|y_1|} - \sqrt{|y_2|} \\ = \frac{|y_1| - |y_2|}{\sqrt{|y_1|} + \sqrt{|y_2|}}$$

1) $y(x) \equiv 0$ is a solution.

$$2) \quad y(x) = \begin{cases} \frac{x^2}{4}, & x > 0. \\ -\frac{x^2}{4}, & x < 0. \end{cases}$$

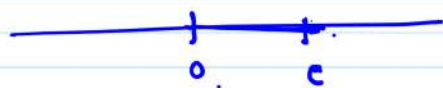
So, f does not satisfy Lipschitz condition.

Example:- $\frac{dy}{dx} = y^{2/3}$, $y(0) = 0$. $\rightarrow$ does not satisfy Lipschitz's condition.
 $y \equiv 0$, is a solution.

$$y^{-2/3} dy = dx.$$

$$\frac{y^{-2/3+1}}{-2/3+1} = x.$$

$$\Rightarrow \frac{y^{1/3}}{3} = x \Rightarrow y = \frac{x^3}{27}.$$



Example 3. $\frac{dy}{dx} = y^{2/3} + 1$, $y(0) = 0$. $\rightarrow$ does not satisfy Lipschitz's condition.
 has a unique solution.
 $y^{1/3} = z$.

Remark 1- Lipschitz's condition is sufficient but not necessary condition.

§ Picard iteration for initial value problems

IVP: $\frac{dy}{dx} = f(x, y)$ $y(x_0) = y_0$.

$$y_0(x) = y_0.$$

$$y_1(x) = y_0 + \int_{x_0}^x f(t, y_0(t)) dt.$$

$$y_2(x) = y_0 + \int_{x_0}^x f(t, y_1(t)) dt.$$

$$y_n(x) = y_0 + \int_{x_0}^x f(t, y_{n-1}(t)) dt.$$

We are getting sequence of function $\{y_n(x)\}$.

Theorem: (convergence of Picard iteration).

Under the hypothesis of Existence theorem and

Theorem*. $\{y_n(x)\}$ converges to a unique.

Solution. $y = y(x)$ defined in $|x - x_0| < \alpha$.
Where, $\alpha = \min\{a, \frac{b}{K}\}$.

Example. $y' = 1 + y^2$, $y(0) = 0$.

$$y_0(x) = 0, \quad y_1(x) = x, \quad y_2(x) = x + \frac{x^3}{3}$$

$$, \quad y_3(x) = x + \frac{x^3}{2} + \frac{2}{15} x^5 + \frac{1}{16} x^7.$$

$$y = \tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

Example. $y' = xy$, $y(0) = 1$.

$$y_0(x) = 1, \quad y_1(x) = 1 + \frac{x^2}{2}, \quad y_2(x) = 1 + \frac{x^2}{2} + \frac{(\frac{x^2}{2})^2}{2!}$$

$$y = e^{x^2/2}, \quad x \in \mathbb{R}.$$

Concluding remark:-

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

$f(x, y) = \frac{2(y-1)}{x}$, $x_0 = 0$, $y(x_0) = 0$.
 discontinuous and solution does not exist.

$$\frac{dy}{dx} = f(x, y), \quad y(0) = 1.$$


 discontinuous.

$y(x) = 1 + cx^2$, $c \rightarrow$ arbitrary constant.
 infinitely many solution.